User Acceptance Testing (UAT) Test Cases — WariMitra

Document Identifier: WM-UAT-TC-017

Classification: Operational Acceptance Scripts & Test Execution Matrix

Standard: IEEE Std 829-2008 / Traceable to `IEEE_TEST_CASES.md`

Target Audience: Business Analysts, QA Test Leads, UAT Stakeholder Sign-Off Committees

Execution Environment: Staging Environment (`warimitra-staging` / `asia-south1`)

1. Test Case Execution Matrix

Test Case ID	Req ID	Scenario	Preconditions	Test Data	Test Steps	Expected Result	Actual Result	Status	Business Owner	Defect Ref
UAT-SOS-01	`FR-SOS-001`, `BR-001`	Validate 3-second hold SOS emergency panic dispatch	User logged in on Android device with GPS enabled.	Lat: `18.5204`, Lon: `73.8567`, Severity: `critical`	1. Open mobile app to SOS screen. 2. Press and hold red button for exactly 3.0	Radial progress completes; haptic buzz triggers; `createSOS` creates document in `sosAlerts`;	Alert created in 1.2s; push received on control room screen.	▨ **PASS**	Lead Police Inspector	None

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					seconds. 3. Observe UI transition.	screen displays help confirmed with 108 dial button.				
UAT-SOS-02	FR-SOS-001	Verify early release cancels emergency dispatch	User on SOS screen.	Release at 1.5 seconds	1. Press red SOS button. 2. Release finger at 1.5 seconds.	Radial progress resets to zero; no network request is sent; no alert appears in contro	Button reset immediately; zero database write.	✅ **PASS**	Lead Police Inspector	None

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						contro l room.				
UAT-SOS-03	`FR-SOS-002`, `BRULE-001`	Verify duplicate dispatch prevention via idempotency	Flaky network simulation; identical idempotency key.	`idempotencyKey`: `uuid-uat-1001`	1. Send first `createSOS` request. 2. Immediately re-send identical request with same key.	First call creates `sosAlerts` doc; second call returns existing doc ID without creating second ticket or second FCM alert.	Handled identically; no duplicate incident card created.	◍ **PASS**	Solution Architect	None
UAT-	`FR-`	Valida	User	6 calls	1.	First 5	Cloud	◍	Syste	None

UAT-	FR-	Valida	User	6 calls	1.	First 5	Cloud		Syste	

Test Case ID	Req ID	Scenario	Preconditions	Test Data	Test Steps	Expected Result	Actual Result	Status	Business Owner	Defect Ref
SOS-04	SOS-003, BRULE-002	te rate limiting lockout at 6th alert	account active.	within 45 seconds	Invoke createSOS 5 times in rapid succession. 2. Attempt 6th invocation within same minute.	succeeded; 6th invocation fails with resource-exhausted; app displays Marathi coold own dialog.	Function threw 429; user shown wait message.	**PASS**	m Administrator	
UAT-SOS-05	FR-SOS-004, BRULE-003	Verify rejection of coordinates outside India	Simulated mock GPS location.	Lat: 51.5074 (London), Lon: -0.1278	1. Submit createSOS with London coordi	Cloud Function rejects with failed-precondition	Function returned error; no record in	**PASS**	Lead Police Inspector	None

Test Case ID	Req ID	Scenario	Preconditions	Test Data	Test Steps	Expected Result	Actual Result	Status	Business Owner	Defect Ref
					coordinates.	nditi on; request rejected.	in sosAlerts.			
UAT-MED-01	FR-MED-002, BR-002	Discover active medical camps within 10 km radius	User in Saswad area; 3 camps seeded (2km, 7km, 15km).	Lat: 18.3448, Lon: 74.0298, Radius: 10km	1. Open Medical tab in mobile app. 2. Tap "Nearby Medical Camps".	Returns 2 camps (2km and 7km) sorted by distance; 15km camp omitted; bed availability shown.	2 camps displayed in 340ms; distance labels accurate.	✅ **PASS**	Chief Medical Officer	None
UAT-MED-02	FR-MED-002, BRULE-008	Verify inactive camps are	1 camp marked statu	Camp ID: camp_inactive_0	1. Search medical	Inactive camp is compl	Only active facilities return	✅ **PASS**	Chief Medical Officer	None

Test Case ID	Req ID	Scenario	Preconditions	Test Data	Test Steps	Expected Result	Actual Result	Status	Business Owner	Defect Ref
		are excluded from search	`status: 'inactive'` at 1.5 km.	`ive_01`	al camps within 10 km.	completely excluded from results array.	returned; zero routing to closed camp.			
UAT-CMD-01	`FR-CMD-003`, `BR-003`	Real-time incident receipt without browser refresh	Police operator on Next.js Command Center feed.	New mobile SOS alert triggered in field	1. Operator monitors SOS queue. 2. Field tester triggers SOS on mobile.	New incident card appears dynamically at top of queue in < 2.0 seconds with pulsing red border.	Incident card rendered in 1.4 seconds; audible chime fired.	🟢 **PASS**	Police Control Room	None
UAT-CMD-	`FR-CMD-0`	Dispatcher	Operator	Incident ID:	1. Click	Incident	Status succes	🟢 **PASS**	Police Contr	None

CMD-	CMD-0	cher	tor	nt ID:	Click	nt	succes		Contr	

Test Case ID	Req ID	Scenario	Preconditions	Test Data	Test Steps	Expected Result	Actual Result	Status	Business Owner	Defect Ref
02	03, BRULE-005	acknowledges and resolves incident	authenticated as police_officer.	sos_uat_test_01	"Acknowledge". 2. Select "Dispatch Unit". 3. Enter notes and click "Resolve".	transitions status: pending $\rightarrow$ acknowledged $\rightarrow$ dispatched $\rightarrow$ resolved; UI updates across all clients.	sfully updated; removed from active triage queue.		ol Room	

Test Case ID	Req ID	Scenario	Preconditions	Test Data	Test Steps	Expected Result	Actual Result	Status	Business Owner	Defect Ref
UAT-W3-01	FR-W3-004, BR-004	Verify cryptographic blockchain proof generation	Sepolia testnet configured; SOS incident resolved.	Incident ID: sos_uat_test_01	1. Check blockchainProofs collection for entityId. 2. Inspect transaction hash and SHA-256 canonical hash.	Document exists containing valid 64-char SHA-256 hash and Sepolia transaction hash; status CONFIRMED.	Proof confirmed on Sepolia testnet; verified via block explorer.	**PASS**	District Magistrate	None
UAT-FAM-01	FR-007	Create and join family group via 6-	2 test phones running mobil	Code: WARI26	1. Phone 1 creates family	Phone 2 joins circle; both devices	Both pins rendered accurately	**PASS**	Pilgrim Representative	None

		via 6-char code	mobile app.		family circle "Jadhav Family". 2. Phone 2 enters code `WARI26`.	s appear on shared map with distance and battery level.	tely on map.			
UAT-MIS-01	`FR-009`	Missing person intake with photo upload	Test device with camera.	Name: "Anand Shinde", Age: 72, Photo: 3.5MB JPEG	1. Fill missing person form. 2. Capture photograph. 3. Submit report.	Photo uploaded to Cloud Storage in < 5s; case document created in `missingPersons`;	Upload took 3.2s; photo rendered in police dashboard.	▨ **PASS**	Police Crime Branch	None

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						visible on police portal.				

2. Summary of UAT Results

- **Total Test Cases Executed:** 12
- **Passed:** 12 (100%)
- **Failed:** 0 (0%)
- **Blocked:** 0 (0%)
- **UAT Sign-off Recommendation: APPROVED FOR PILGRIMAGE GO-LIVE**